Thinking Moves

Twelve five-minute thinking tasks, each built on the topic you are teaching. AI can prepare a new task in seconds; your students do the thinking without devices.

Five minutes in the lesson

Task · 2 min
Students work alone, on paper.

Pair talk · 1 min
Compare and improve one answer.

Share · 1 min
Take two or three answers; ask the Probe.

Think back · 1 min
Each student finishes the sentence starter.

Six kinds of thinking

1. Generating ideas

Many ways

2. Evaluating and choosing

Pick and justify, Check the claim, Argue it

3. Comparing and transferring

Like what?

4. Explaining and reasoning from basics

Best explanation, Strip it down, Estimate it, Model it

5. Causes and systems

Then what?, Map the loop

6. Regulating your own thinking

Think back

How strong is the evidence?

Strong Meta-analyses of classroom interventions show that teaching this kind of thinking improves it.

Promising Controlled studies show gains, but results are mixed, small in number or dependent on context.

Framework A useful structure for thinking, with no intervention studies behind it yet.

Labels rate the research on teaching this kind of thinking regularly, not a single five-minute move. One move is one small piece of that practice.

MOVE 1 · GENERATING IDEAS

Many ways

Students produce as many different solutions as they can, then look for the most unusual.

Promising Divergent, associative and lateral thinking · 5 minutes · no devices

Technology · Grade 8

TASK	Find as many ways as you can to keep a packed lunch cold until noon without a fridge or electricity. Aim for ten. Then circle the two most unusual ones.
LOOK FOR	Ideas that use different principles, not ten versions of one: insulation (wool, foil, a vacuum flask), cold mass (a frozen bottle, an ice pack), evaporation (a wet cloth), shade and position, and changing the food itself (packing it frozen).
PROBE	Which principle does each idea use? Is there a principle nobody in the class has used yet?
THINK BACK	My best idea came after I stopped thinking about …

Music · Grade 10

TASK	Take a melody everyone knows, such as "Happy Birthday". List as many ways as you can to make it sound sad without changing a single note of the melody.
LOOK FOR	Ideas across several musical elements: tempo, dynamics, instrument and timbre, minor chords in the accompaniment under the same melody, articulation, the rhythm of the accompaniment, and the setting it is played in.
PROBE	Which musical element did nobody change? Try one idea from that element.
THINK BACK	I got stuck on … until I switched to thinking about …

Evidence. Creativity training works on average (Haase, Hanel & Gronau 2023: g = 0.53), but that meta-analysis covers adults, and the evidence for school-age students is older and less certain. Labels rate research on teaching this kind of thinking regularly, not a single move.

MOVE 2 · EVALUATING AND CHOOSING

Pick and justify

Students agree on criteria first, then choose one option and defend it against each criterion.

Promising · Convergent thinking · 5 minutes · no devices

Geography · Grade 9

TASK	A town must choose one of three sites for a wind farm: a hilltop beside a nature reserve, flat farmland 15 km from the nearest road, or a coastal strip next to a holiday village. Agree on three criteria first, then pick a site and justify it against each criterion.
LOOK FOR	Criteria set before the choice, such as wind exposure, the cost of roads and grid connection, and the impact on wildlife and people. Strong answers admit what their choice costs and say which criterion they weighted most.
PROBE	Which criterion would have to change for you to pick a different site?
THINK BACK	The criterion I almost forgot was …

History · Grade 10

TASK	You want to know how ordinary Londoners felt during the Blitz in 1940. You have three sources: a private diary written that week, a newspaper report from the next morning, and a school textbook from 2010. Rank them for this question and justify the order.
LOOK FOR	A ranking tied to the question: the diary is closest to private feelings but shows one person; the wartime newspaper was shaped by censorship and the aim of keeping up morale; the textbook summarises with hindsight. Strong answers say what each source can and cannot show.
PROBE	Would your ranking change if the question were "How many bombs fell on London?"
THINK BACK	I changed my ranking when I noticed …

Evidence. In the Haase, Hanel & Gronau 2023 meta-analysis, training convergent thinking showed larger effects than training divergent thinking, but the studies are on adults. In school it is part of ordinary problem solving. Labels rate research on teaching this kind of thinking regularly, not a single move.

MOVE 3 · EVALUATING AND CHOOSING

Check the claim

Before deciding whether a claim is true, students say what evidence would convince them and what would not.

Strong · Critical thinking · 5 minutes · no devices

Biology · Grade 10

TASK	Claim: "Cracking your knuckles gives you arthritis." Before deciding whether it is true, write down what kind of evidence would convince you, and what kind would not.
LOOK FOR	The difference between an anecdote ("my grandad did it and has arthritis") and a comparison of people who crack their knuckles with people who don't. Strong answers ask for a comparison group, a large sample and years of follow-up. Studies comparing the two groups have not found more arthritis among knuckle-crackers.
PROBE	One doctor cracked the knuckles of one hand for more than 50 years and never the other. What does that show, and what can it not show?
THINK BACK	The evidence I would have accepted too quickly was …

English · Grade 11

TASK	An advert says: "80% of dentists recommend our toothpaste." List every question you would need answered before believing what the advert wants you to think.
LOOK FOR	Questions about who was asked, how many, the exact wording of the question, whether dentists could name more than one brand, and who paid for the survey. In 2007 the UK advertising regulator ruled a claim like this misleading because the dentists surveyed could recommend several brands.
PROBE	Rewrite the advert so it says only what such a survey could prove.
THINK BACK	The question I would not have thought of on my own was …

Evidence. Explicit teaching of critical thinking improves it across subjects and ages (Abrami et al. 2015: g = 0.30 over 341 effect sizes), most when combined with discussion and authentic problems. Labels rate research on teaching this kind of thinking regularly, not a single move.



MOVE 4 · EVALUATING AND CHOOSING

Argue it

Students build an argument with a claim, evidence and the reasoning that links them, and answer the other side.

Promising · Argumentation from evidence · 5 minutes · no devices

Civics · Grade 11

TASK	Claim: "The voting age should be lowered to 16." Write one argument for and one against. Each needs a claim, a piece of evidence or an example, and the reasoning that links them.
LOOK FOR	Evidence, not just opinion. For example, Austria lowered the voting age to 16 for national elections in 2007, so there is real experience to point to. Strong answers explain why the evidence supports the claim and answer the strongest argument on the other side.
PROBE	What evidence would make you change sides?
THINK BACK	My argument got stronger when ...

Chemistry · Grade 10

TASK	Claim: "Recycling an aluminium can saves more than recycling a glass bottle." Decide whether you agree and argue it with a claim, evidence and reasoning. Say what "saves" means in your argument.
LOOK FOR	A definition of "saves" (energy, raw material or emissions). Key evidence: making aluminium from recycled metal uses about 95% less energy than making it from ore, because the electrolysis step is skipped; recycled glass saves far less, since it has to be melted either way. Strong answers say what the claim leaves out, such as weight and transport.
PROBE	Which part of your reasoning depends on a number you would want to check?
THINK BACK	Defining ... first changed my answer.

Evidence. School interventions that teach scientific reasoning and argumentation show a medium effect on average across 30 interventions (Engelmann, Neuhaus & Fischer 2016). Labels rate research on teaching this kind of thinking regularly, not a single move.

Like what?

Students set a new idea beside a familiar one, map what matches, and find where the comparison breaks.

Strong · Analogical thinking · 5 minutes · no devices

Physics · Grade 9

TASK	An electric circuit is often compared to water flowing in pipes. Fill in: battery = ?, wire = ?, current = ?, resistance = ?. Then find at least two places where the comparison breaks.
LOOK FOR	Battery as a pump, current as the flow, resistance as a narrow section of pipe. Where it breaks: cut a pipe and water pours out, but cut a wire and the current simply stops; electrons in a wire drift slowly even though a lamp lights almost instantly. Strong answers say what the comparison is good for and when to stop using it.
PROBE	Can you find a better comparison for the part where water pipes fail?
THINK BACK	The comparison helped me until …

Religion and ethics · Grade 10

TASK	Is a promise like a contract? List what they have in common, then where the comparison breaks down. Give one example of each.
LOOK FOR	Both create obligations and expectations. Where it breaks: a contract can be enforced by a court and a promise usually cannot; a contract needs both sides to agree, while a promise can be one-sided; breaking a promise is judged morally rather than legally. Strong answers ask what the comparison hides.
PROBE	Which kind of promise comes closest to being a contract? Why?
THINK BACK	The difference I did not notice at first was …

Evidence. Learning by comparing cases improves understanding across 57 experiments (Alfieri, Nokes-Malach & Schunn 2013), and comparing solution methods built flexibility in Grade 7–8 algebra (Rittle-Johnson & Star 2007). Labels rate research on teaching this kind of thinking regularly, not a single move.



Best explanation

Students meet a puzzling observation, propose several explanations, and say what evidence would tell them apart.

Promising · Abductive thinking · 5 minutes · no devices

Biology · Grade 9

TASK	The bean plants on the windowsill all lean toward the glass. The ones on the table in the middle of the room grow straight up. Write three possible explanations. For each, name one thing you could observe that would support it.
LOOK FOR	Light from one side (phototropism), plus at least one real alternative such as warmth or a draught, each with a test that separates it from the others.
PROBE	Which explanation would still stand if we turned the pots every day?
THINK BACK	The explanation I gave up first was … because …

History · Grade 11

TASK	Archaeologists find an English village that was lived in for 300 years and empty by 1500. Give three explanations, and the evidence that would make each one more likely.
LOOK FOR	Plague (the population fell sharply after 1349, though villages rarely emptied from plague alone; look for burials and tax records), land turned into sheep pasture (gradual, mostly in the 1400s; enclosure records and new field boundaries), and failing soil or climate. Strong answers match each cause with a kind of evidence.
PROBE	Which single piece of evidence could rule out two explanations at once?
THINK BACK	I changed my mind when …

Evidence. Generating and testing hypotheses is part of scientific reasoning, which school interventions improve on average (Engelmann, Neuhaus & Fischer 2016). Studies of abductive reasoning on its own are few and small. Labels rate research on teaching this kind of thinking regularly, not a single move.

Strip it down

Students set aside the rule they were given and rebuild it from what they know for certain.

Framework · First-principles thinking · 5 minutes · no devices

Maths · Grade 10

TASK	Without using the formula, show why the area of any triangle is half of base times height. Start only from what you know for certain: the area of a rectangle.
LOOK FOR	A triangle drawn inside a rectangle with the same base and height, split along its height into two right-angled triangles, each half of a smaller rectangle. Strong answers check the case where the height falls outside the triangle.
PROBE	Does your argument still work if the triangle has an angle greater than 90°? What would you have to change?
THINK BACK	The step I could not justify at first was …

Art · Grade 8

TASK	Forget what you have been taught about drawing in 3D. On a flat page, what are the basic things that make one object look closer than another? List them, then test each one with a quick sketch.
LOOK FOR	Overlap, size, height on the page, detail and sharpness, colour and contrast fading with distance, lines that converge, shading and cast shadows. Strong answers test each cue on its own.
PROBE	Which cue is strong enough to work on its own, without any of the others?
THINK BACK	I found the basics by asking …

Evidence. There are no intervention studies of first-principles thinking itself. The closest research is productive failure, where students try to build a method before being taught it. Across 53 studies it helped conceptual understanding more than teaching first, but not procedural skill (Sinha & Kapur 2021). Labels rate research on teaching this kind of thinking regularly, not a single move.

Estimate it

Students reach a rough answer to a big question by breaking it into steps they can guess, and saying every assumption out loud.

Framework · Estimation (Fermi problems) · 5 minutes · no devices

Maths · Grade 9

TASK	Roughly how many times will your heart beat in a lifetime? Work in pairs, write down every assumption and round freely. Aim for the right power of ten, not an exact answer.
LOOK FOR	A chain such as 70 beats a minute × 60 × 24 × 365 × 80 years, which gives about 3 billion. Strong answers state their assumptions, round sensibly and check whether the result is plausible.
PROBE	Which assumption changes the answer most if you get it wrong?
THINK BACK	Writing my assumptions down helped me notice …

PE and health · Grade 10

TASK	Estimate how many kilometres a professional football midfielder runs in a whole league season. Show each step and the assumption behind it.
LOOK FOR	A distance per match (top-level outfield players often cover around 10–12 km), the number of league matches (38 in a 20-team league), and a decision about cup games and training. Strong answers say which number they are least sure of.
PROBE	Would training add more distance than matches? How could you find out?
THINK BACK	My estimate changed most when I …

Evidence. Estimation carries real weight in maths and science curricula, but research on Fermi problems mostly describes how students solve them (Albarracín & Ärlebäck 2019) rather than testing whether they improve students' estimation. Labels rate research on teaching this kind of thinking regularly, not a single move.



MOVE 9 · EXPLAINING AND REASONING FROM BASICS

Model it

Students turn a real situation into a simple mathematical model, use it, and say where it stops matching reality.

Promising · Mathematical modelling · 5 minutes · no devices

Maths · Grade 11

TASK	A rumour starts with one student. Every hour, everyone who knows it tells one new person. How many know after 1, 2, 3 … hours? Then think about a school of 600 students and change your model to fit.
LOOK FOR	Doubling at first (1, 2, 4, 8 … passing 600 after about 10 hours), then the realisation that the spread must slow down as fewer people are left to tell, giving an S-shaped curve. Strong answers state the model's assumptions and where it stops being realistic.
PROBE	Which assumption in your model is least like real life?
THINK BACK	My first model broke when …

Economics · Grade 12

TASK	The school concert has 300 seats. At €5 a ticket, 300 people come; for every €1 more, 30 fewer people come. Build a model of the ticket income and find the best price. Then name one thing the model ignores.
LOOK FOR	Income = price × audience. The best whole-euro price is €7 or €8 (both give €1,680); €7.50 gives €1,687.50. Strong answers question the assumption that the audience falls by the same amount for every euro, and ask what the concert is for: money or a full hall.
PROBE	If the goal were a full hall instead of the most money, what would your model recommend?
THINK BACK	The assumption I accepted without checking was …

Evidence. Teaching mathematical modelling improved problem solving and achievement in a meta-analysis of 20 studies (Hidayat et al. 2025). The field is small. Labels rate research on teaching this kind of thinking regularly, not a single move.

Then what?

Students follow a decision through its first consequence, the consequence of that, and one more.

Framework · Second-order and causal thinking · 5 minutes · no devices

History · Grade 10

TASK Hanoi, 1902. To fight rats that spread plague, the authorities pay a reward for every rat tail handed in. Write the first consequence, then the consequence of that, then one more.

LOOK FOR More than "people kill more rats": people cut off tails and let the rats go free to breed, and some started breeding rats for the reward, as the historian Michael Vann describes. Strong answers trace how the reward changed what people wanted.

PROBE How would you design the reward so the chain does not go wrong?

THINK BACK The consequence I missed at first was ...

Geography · Grade 12

TASK In 1989 Mexico City banned each car from driving on one weekday, based on its number plate, to cut air pollution. Write the chain of consequences: first, second, third.

LOOK FOR First, fewer cars on the road. Then some households bought a second, often older car to drive on their banned day. The economist Lucas Davis found no evidence that the ban improved air quality. Strong answers ask how people adapt to a rule.

PROBE What would you have to measure to know whether the ban worked?

THINK BACK I started thinking further ahead when ...

Evidence. Teaching students to see patterns of cause and effect has been developed through design research (Grotzer, Harvard Project Zero), but there are no effect sizes for it yet. Labels rate research on teaching this kind of thinking regularly, not a single move.



MOVE 11 · CAUSES AND SYSTEMS

Map the loop

Students draw how the parts of a system affect each other, and find the loops that make it grow or settle.

Promising · Systems thinking · 5 minutes · no devices

Biology · Grade 10

TASK	Draw the connections between sea otters, sea urchins and kelp forests on the Pacific coast. Then show what happens to all three if the otters are hunted out.
LOOK FOR	Otters eat urchins and urchins eat kelp: fewer otters means more urchins and less kelp. Estes and Palmisano showed this in 1974 by comparing islands with and without otters. Strong answers add a link back, such as the kelp sheltering fish and other prey.
PROBE	If the otters returned, what would happen first, and what would take years?
THINK BACK	Drawing it showed me …

Social studies · Grade 9

TASK	A new café opens near the school and becomes popular. Draw a loop with three parts: how popular it is, how long the queue gets, and how many customers give up and go elsewhere. Does the loop keep growing or balance itself?
LOOK FOR	More popular, longer queue, more people give up, fewer customers, shorter queue: a loop that balances itself. Strong answers add a loop that grows (crowds attract more customers) and say which one wins.
PROBE	What could the café owner change to break the balancing loop?
THINK BACK	The arrow I nearly drew the wrong way was …

Evidence. About 50 Grade 8 students taught with an earth-systems approach progressed through levels of systems thinking, and a third reached the highest level (Ben-Zvi Assaraf & Orion 2005). There is no meta-analysis yet, and few secondary-school trials. Labels rate research on teaching this kind of thinking regularly, not a single move.

MOVE 12 · REGULATING YOUR OWN THINKING

Think back

Students describe how they thought, not what they found, and decide what to do differently. It closes every move and works on its own.

Strong · Metacognitive thinking · 5 minutes · no devices

Modern languages · Grade 8

TASK	Right after the vocabulary quiz, before you see your score, write: How did you study? Which words do you think you got wrong, and why those? Then compare with your result.
LOOK FOR	Specific descriptions of how they studied (reading the list or testing themselves), predictions about which words they missed, and an honest comparison afterwards. Strong answers name one change for next time.
PROBE	Was your prediction better or worse than you expected? What does that tell you?
THINK BACK	Next time I will study by … instead of …

Maths · Grade 12

TASK	After solving a hard problem, write three lines: where I got stuck, what I tried, and what finally moved me forward. Then swap with a partner and compare strategies.
LOOK FOR	Named strategies, such as drawing a diagram, trying a simpler case or working backwards, rather than "I just thought harder". Strong answers notice a strategy they could use on the next problem.
PROBE	Which of your partner's strategies would have saved you time?
THINK BACK	When I get stuck next time, I will first try …

Evidence. Teaching students to plan, monitor and evaluate their own learning is one of the strongest strands in the EEF Teaching and Learning Toolkit: +8 months of progress on average, with 4 of 5 padlocks for the strength of the evidence. Labels rate research on teaching this kind of thinking regularly, not a single move.

Your own task

Fill in the blanks, then type the prompt into any AI tool: Copilot, Gemini, ChatGPT or Claude. Check the answer before the lesson, especially anything under CHECK.

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Do not include student names or student work in this conversation.

You are helping a Grade _____ _____ teacher plan a
five-minute thinking task for tomorrow's lesson.

Thinking move: _____ (copy the move's name and
one-line description from its card)

Lesson topic: _____
Students already know: _____

Write the task so it can only be done by thinking about this
topic's content, and so students need no devices: paper,
whiteboards or talk.

Reply in plain text, no tables, with exactly these six parts:
MOVE: the move's name and one line on what it trains
TASK: the task, written to the students: at most three
  short steps, under 80 words, about two minutes of work
LOOK FOR: at most five short bullets: the strongest answer
  first, then the most common mistake
PROBE: one follow-up question that pushes past the first answer
THINK BACK: one sentence starter about how they thought
CHECK: every fact, figure, date or source to verify,
  one per line, or None
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Faster online: the prompt builder at choosewise.education/thinking-moves fills this in for you. Students do not need AI for any move. If older students use AI themselves, check the tool's age rules, your national rules and your school policy first.